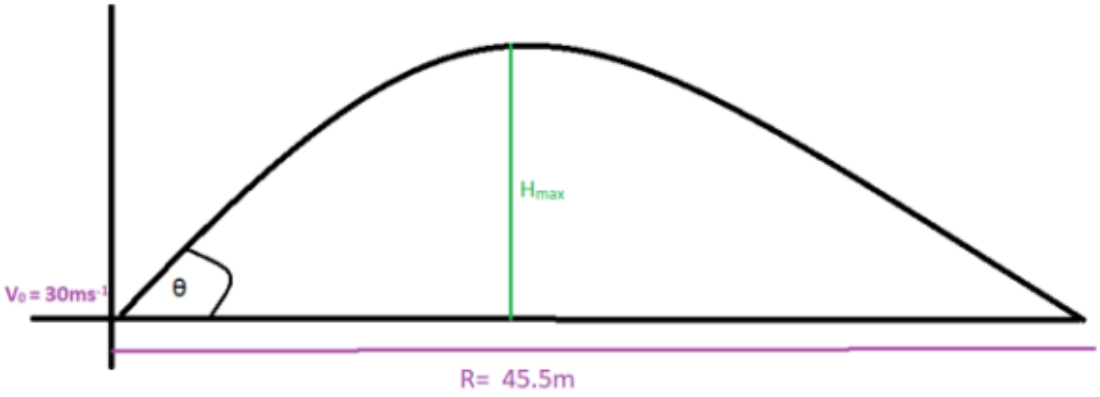
MTH404 Dynamics Fall 2023
Assignment No.2

Question No1:



Solution:

Given Data:

$v_o = 30ms^{-1}$

$R = 45.5m$

$g = 10ms^{-2}$

Find out:

a. The angle of θ (in degree) = ?

b. The time of Flight T (in second) = ?

c. The maximum height $H_{\max}$ that the particle will attain (in meters) = ?

Calculation:

a. The angle of θ (in degree)

$$R = \frac{v_o^{\,2}\sin 2\theta}{g}$$

$putting\ the\ values$

$$45.5m = \frac{(30ms^{-1})^2 \sin 2\theta}{10ms^{-2}}$$

$$45.5m \times 10ms^{-2} = (900m^2\,s^{-2})\sin 2\theta$$

$$\frac{455m^2s^{-2}}{900m^2\,s^{-2}} = \sin 2\theta$$

$$\Rightarrow \sin 2\theta = 0.5056$$

$$2\theta = \sin^{-1}\left(0.5056\right)$$

$$2\theta = 30.37^{\circ}$$

$$\theta = \frac{30.37^{\circ}}{2} = 15.19^{\circ}$$

b. The time of Flight T (in second)

$$T = \frac{2v_o \sin\theta}{g}$$

$$T = \frac{2(30ms^{-1})\sin(15.19)}{10ms^{-2}}$$

$$T = \frac{(60)(0.2620)}{10}s$$

$$T = 1.57s \approx 1.6s$$

c. The maximum height $H_{\max}$ ***that the particle will attain (in meters)***

$$H_{\max} = \frac{{v_o}^2 \sin^2\theta}{2g}$$

$$H_{\max} = \frac{(30ms^{-1})^2 \sin^2\left(15.19\right)}{2(10ms^{-2})}$$

$$H_{\max} = \frac{(900)(0.2620)^2}{20}m$$

$$H_{\max} = \frac{(900)(0.0686)}{20}m$$

$$H_{\max} = 3.087m$$